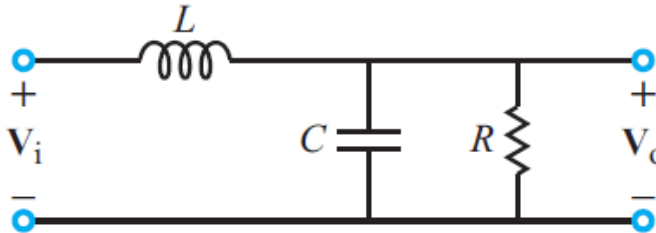
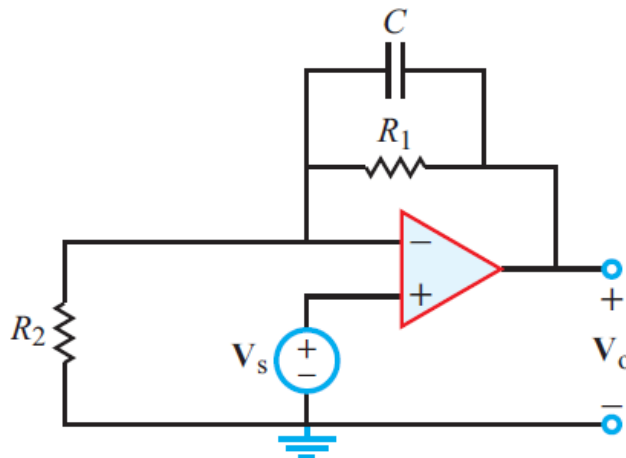


1. (25 points) For the following circuit, derive an expression for $\mathbf{H}(\omega)$ and plot the magnitude and phase of $\mathbf{H}(\omega)$ for $R = 100\Omega$, $C = 1\mu F$, $L = 0.1mH$ (you may use MATLAB).



2. (25 points) For the following circuit, derive an expression for $\mathbf{H}(\omega)$ and plot the magnitude and phase of $\mathbf{H}(\omega)$ for $R_1 = 99k\Omega$, $R_2 = 1k\Omega$, $C_1 = 0.1\mu F$ (you may use MATLAB). What type of filter is it?



3. (25 points) An audio signal $x(t)$ that is bandlimited to 20kHz is sampled at 25 kHz.
 - a. What portion of its spectrum can still be recovered perfectly from its samples, assuming that you have access to an ideal low-pass filter, and
 - b. What is the cutoff frequency for this ideal low-pass filter?
4. (25 points) An audio signal $x(t)$ is bandlimited to 20kHz.
 - a. Using the convolution property in the frequency domain, calculate the bandwidth of the signal $y(t) = x^2(2t - 1)$
 - b. Calculate the Nyquist sampling rate for the signal $y(t)$